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SET 09 :

1. Import the dataset in R. Plot a histogram for Quiz_Score. Draw a boxplot of Quiz_Score by Course.

Add title and labels. Interpret student performance.

2. Create a scatter plot using R programming between Study_Time and Quiz_Score. Represent

Videos_Watched using bubble size. Apply transparency. Explain the learning pattern.

3. Using R programming, convert Login_Date into Date format. Calculate average quiz score per month. Plot a line chart. Apply moving average smoothing. Identify the trend.

4. Import Online_Learning_Activity.csv into Tableau. Create a bar chart showing average quiz score

by Course. Create a scatter plot (Study Time vs Quiz Score). Add filter for Gender and Course.

Design a mini interactive dashboard.

SOLUTION :

CODE :

01.

```
%%R
```

```
library(ggplot2)
```

```
df9 <- data.frame(
```

```
  Student_ID = c("L01", "L02", "L03", "L04", "L05", "L06"),
```

```
  Gender = c("Male", "Female", "Male", "Female", "Male", "Female"),
```

```
  Age = c(20, 22, 19, 21, 23, 20),
```

```
  Course = c("R", "R", "SQL", "R", "R", "SQL"),
```

```
  Study_Time = c(3.5, 4.2, 2.0, 5.0, 2.5, 4.0),
```

```
  Videos_Watched = c(12, 15, 8, 18, 9, 14),
```

```
  Quiz_Score = c(78, 85, 65, 92, 70, 88),
```

```
  Login_Date = as.Date(c("2025-01-05", "2025-01-05", "2025-02-08", "2025-02-08", "2025-03-12", "2025-03-12"))
```

```
)
```

```
p1 <- ggplot(df9, aes(x = Quiz_Score)) +
  geom_histogram(binwidth = 5, fill = "#377EB8", color = "white", alpha = 0.8)
+
  labs(
    title = "Distribution of Quiz Scores",
    x = "Quiz Score",
    y = "Number of Students"
  ) +
  theme_minimal()
```

```
p2 <- ggplot(df9, aes(x = Course, y = Quiz_Score, fill = Course)) +
  geom_boxplot(alpha = 0.7) +
  geom_jitter(width = 0.1, size = 3) +
  labs(
    title = "Quiz Score Distribution by Course",
    x = "Course",
    y = "Quiz Score",
    fill = "Course"
  ) +
  theme_minimal() +
  scale_fill_brewer(palette = "Set2")
```

```
print(p1)
print(p2)
```

02. %%R

```
library(ggplot2)
```

```
df9 <- data.frame(
  Student_ID = c("L01", "L02", "L03", "L04", "L05", "L06"),
  Course = c("R", "R", "SQL", "R", "R", "SQL"),
  Study_Time = c(3.5, 4.2, 2.0, 5.0, 2.5, 4.0),
  Videos_Watched = c(12, 15, 8, 18, 9, 14),
  Quiz_Score = c(78, 85, 65, 92, 70, 88)
)
```

```
ggplot(df9, aes(x = Study_Time, y = Quiz_Score, size = Videos_Watched, color
= Course)) +
  geom_point(alpha = 0.6) +
  scale_size_continuous(range = c(5, 12)) +
  geom_text(aes(label = Student_ID), vjust = -1.2, size = 3.5, show.legend =
```

```
FALSE, color = "black") +
  labs(
    title = "Study Time vs Quiz Score (Bubble Size = Videos Watched)",
    x = "Study Time (Hours)",
    y = "Quiz Score",
    size = "Videos Watched",
    color = "Course"
  ) +
  theme_bw()
```

03 .%%R

```
if (!requireNamespace("zoo", quietly = TRUE)) install.packages("zoo", quiet = TRUE)
```

```
library(ggplot2)
library(dplyr)
library(zoo)
```

```
# Load dataset
df9 <- data.frame(
  Student_ID = c("L01", "L02", "L03", "L04", "L05", "L06"),
  Quiz_Score = c(78, 85, 65, 92, 70, 88),
  Login_Date = as.Date(c("2025-01-05", "2025-01-05", "2025-02-08", "2025-02-08", "2025-03-12", "2025-03-12"))
)
```

```
# Aggregate monthly average
monthly_avg <- df9 %>%
  mutate(Month = format(Login_Date, "%Y-%m")) %>%
  group_by(Month) %>%
  summarise(Avg_Quiz_Score = mean(Quiz_Score))
```

```
# Calculate 2-month rolling mean using zoo
monthly_avg$Moving_Avg <- rollmean(monthly_avg$Avg_Quiz_Score, k = 2,
fill = NA, align = "right")
```

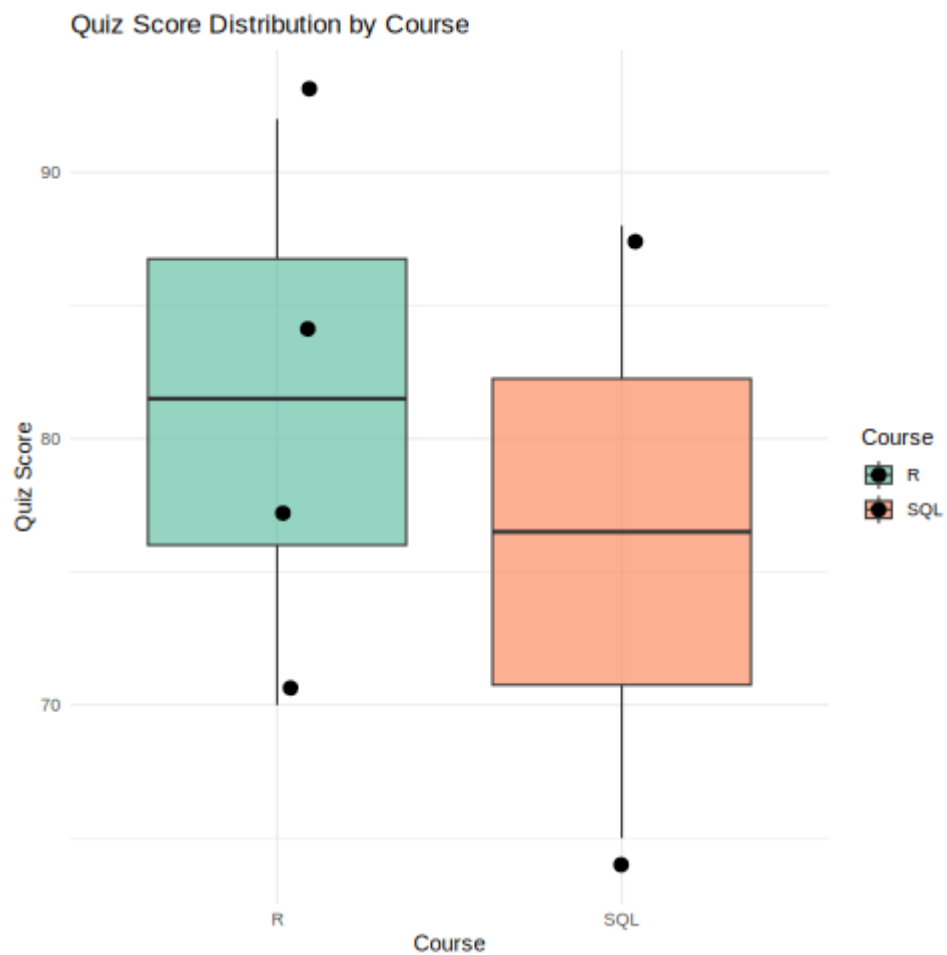
```
# Plot line chart
ggplot(monthly_avg, aes(x = Month, group = 1)) +
  geom_line(aes(y = Avg_Quiz_Score, color = "Monthly Average"), size = 1.2) +
  geom_point(aes(y = Avg_Quiz_Score, color = "Monthly Average"), size = 3) +
  geom_line(aes(y = Moving_Avg, color = "Moving Average"), size = 1.2,
linetype = "dashed") +
  labs(
    title = "Monthly Average Quiz Score Trend with Moving Average",
    x = "Month",
```

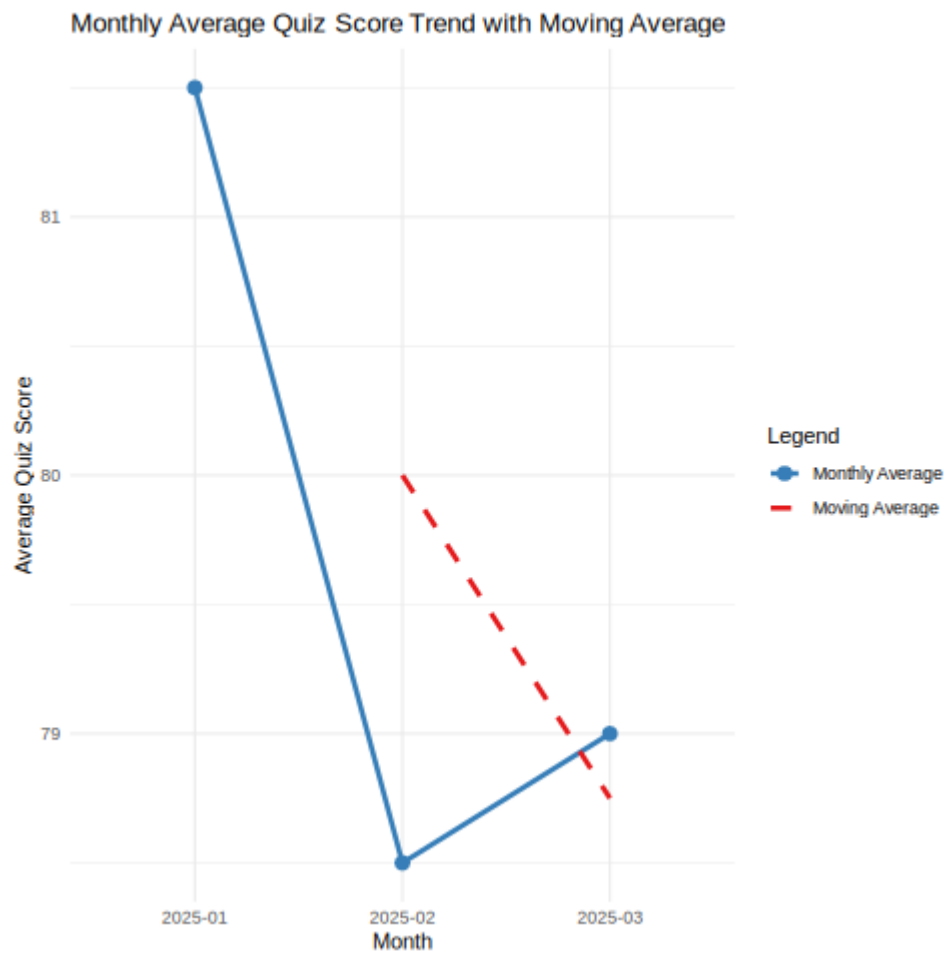
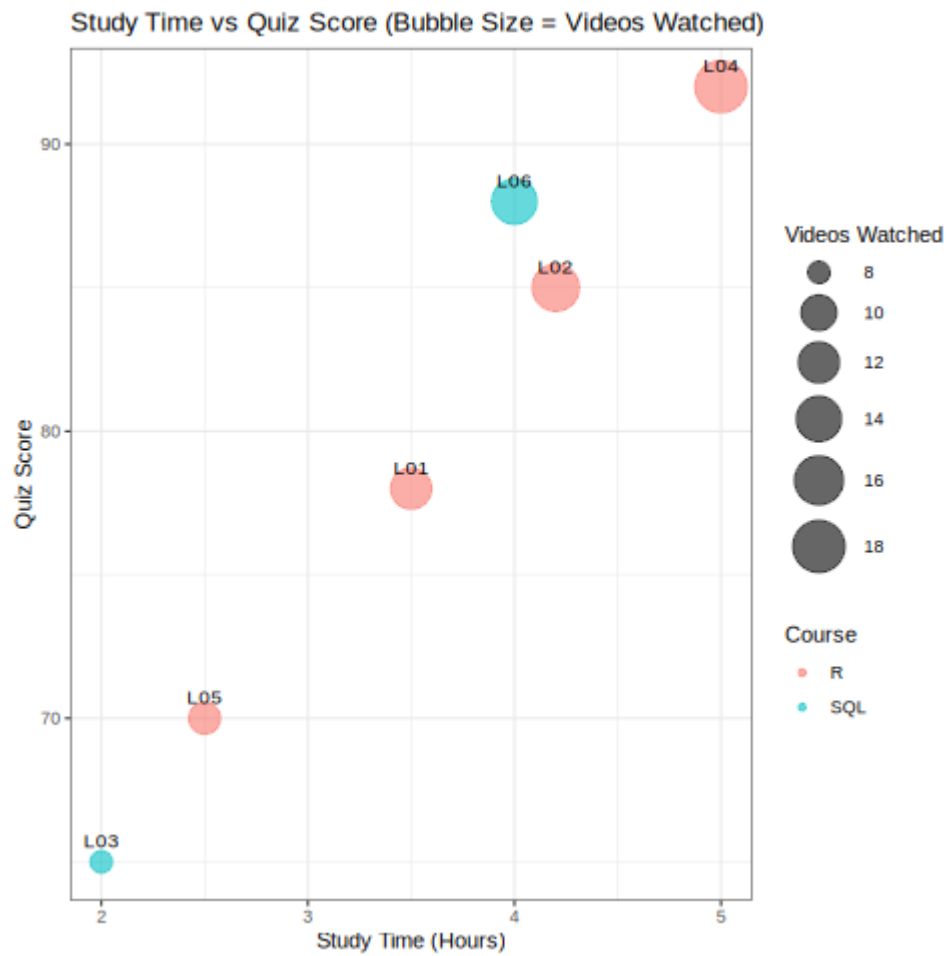
```

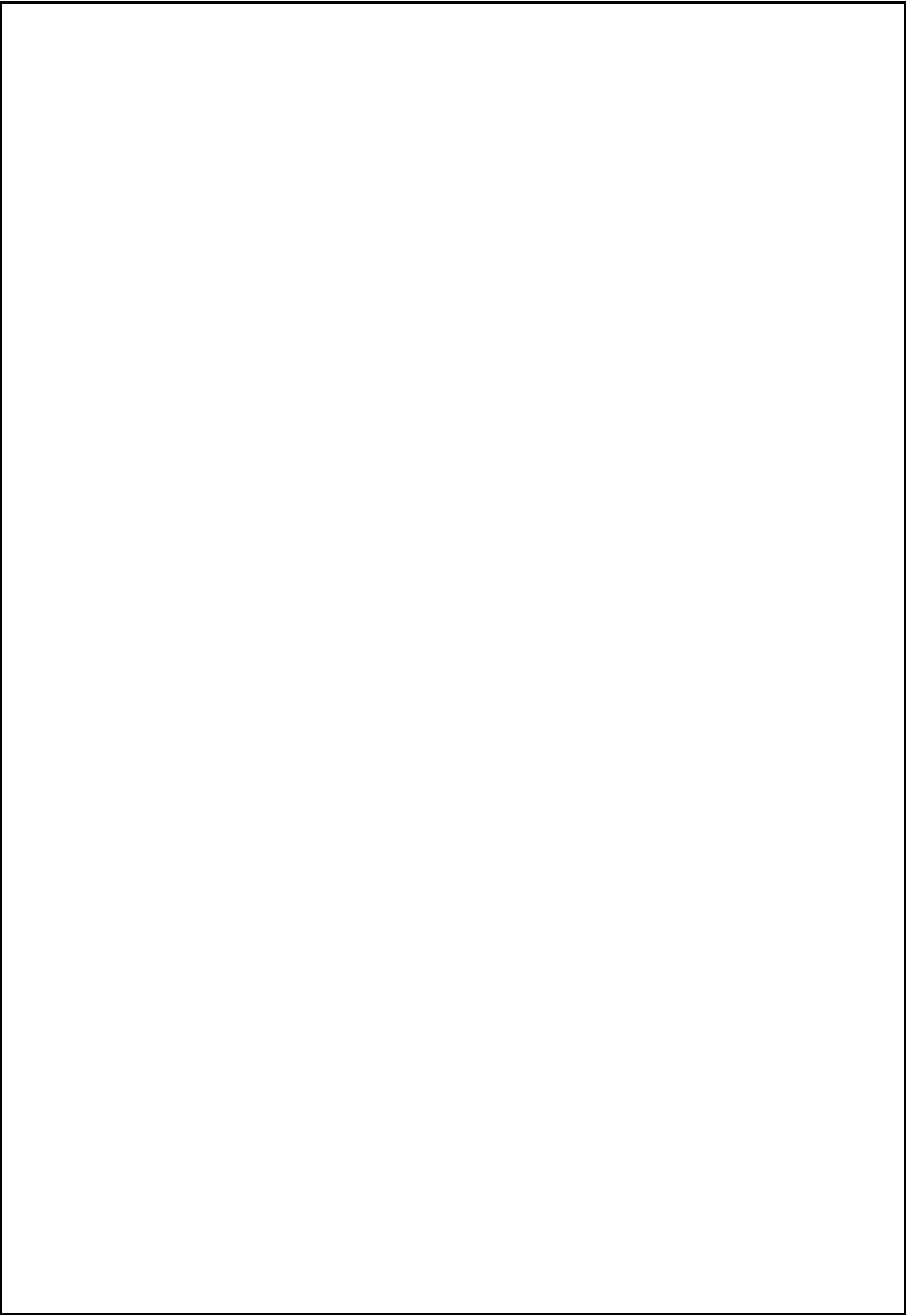
y = "Average Quiz Score",
color = "Legend"
) +
  scale_color_manual(values = c("Monthly Average" = "#377EB8", "Moving
Average" = "#E41A1C")) +
  theme_minimal()

```

OUTPUT :







--_*